Debendra Kumar Swain

Postdoctoral Fellow | TIFR Centre for Applicable Mathematics

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RESEARCH INTERESTS

Numerical analysis of partial differential equations; finite element methods; discontinuous Galerkin methods; computational fluid dynamics.

ACADEMIC APPOINTMENTS

Postdoctoral Fellow - TIFR Centre for Applicable Mathematics
June 2026-present | With Prof. Praveen Chandrashekar

Research Associate - BITS Goa
November 2025-May 2026 | With Prof. Amiya Kumar Pani

EDUCATION

Ph.D. - Indian Institute of Technology Goa, 2019-2025
Thesis: Finite Element Galerkin Approximations to the Boussinesq System of Equations
Supervisor: Prof. Saumya Bajpai | Coursework CGPA: 8.67/10

M.Sc. in Mathematics - Pondicherry University, 2015-2017
CGPA: 8.6/10

B.Sc. (Honours) in Mathematics - Bhadrak Autonomous College, 2012-2015
Overall: 73.66% | Mathematics: 81%

SCHOLARSHIP & ACHIEVEMENT

Joint CSIR-UGC NET in Mathematical Sciences, June 2018. All India Rank: 77.

COMPUTING & LANGUAGES

Python, FEniCS, C++, FreeFem++, MATLAB, ParaView, and LaTeX.
Operating systems: Linux and Windows.
Languages: Odia (mother tongue), English, and Hindi.

JOURNAL PUBLICATIONS

On Error Estimates of a Discontinuous Galerkin Method of the Boussinesq System of Equations
Saumya Bajpai, D. K. Swain.
Computational Methods in Applied Mathematics, 25 (2), 313-348 (2025).
doi:10.1515/cmam-2023-0202

A priori error estimates of a three-step two-level finite element Galerkin method for a 2D-Boussinesq system of equations
Saumya Bajpai, D. K. Swain.
Computers & Mathematics with Applications, 146, 137-164 (2023).
doi:10.1016/j.camwa.2023.06.025

Local discontinuous Galerkin method for Kelvin-Voigt viscoelastic fluid flow model
D. K. Swain, Saumya Bajpai, Deepjyoti Goswami.
Computers & Mathematics with Applications, 200, 467-499 (2025).
doi:10.1016/j.camwa.2025.10.025

ONGOING MANUSCRIPTS

A discontinuous Galerkin method for the 2D-Sobolev equations with Burgers' type nonlinearity.
D. K. Swain, K. Ray, S. Bajpai, and D. Goswami.

On a priori error estimates of a local discontinuous Galerkin method of the 2D-Sobolev equations with Burgers' type nonlinearity.
D. K. Swain.

Discontinuous Galerkin two-grid method for the transient Boussinesq equations.
D. K. Swain and S. Bajpai.

On a priori error estimates of a local discontinuous Galerkin method of the Boussinesq system of equations.
D. K. Swain, S. Bajpai, and D. Goswami.

A priori error estimates for the finite element approximations of 2D-Nematic flow model.
S. Bajpai, A. Majumdar, N. Nataraj, and D. K. Swain.

Manuscripts are listed separately from published articles; submission status is not asserted.

TEACHING EXPERIENCE

Teaching Assistant, IIT Goa
School of Mathematics and Computer Sciences

Linear Algebra and Applications (MA-307) - 2024

Undergraduate Calculus (MTH-101) - 2022, 2023

Ordinary Differential Equations (MA-103) - 2022

Numerical Analysis (MA-213) - 2020, 2021

WORKSHOP TEACHING ASSISTANCE

Partial Differential Equations: Theory and Numerics
IIIT Delhi, 2-14 June 2025.

High-End Workshop on Computational Finance
IIT Goa, 10-16 July 2022.

SELECTED CONFERENCE PRESENTATIONS

LACAM-25, IISER Thiruvananthapuram, 8-11 December 2025
Discontinuous Galerkin two-grid method for the transient Boussinesq equations.

ICAANA 2025, Amrita Vishwa Vidyapeetham, 21-23 July 2025
On error estimates of a discontinuous Galerkin method of the Boussinesq system of equations.

LACAM-24, IISER Thiruvananthapuram, 21-24 February 2024
A priori error estimates of a three-step two-level finite element Galerkin method for a 2D-Boussinesq system of equations.

ACADEMIC TRAINING

Instructional School for Teachers on Partial Differential Equations, Tezpur University, 21 November-3 December 2022.

Advanced Training School on Numerical PDEs and Inverse Problems, IIT Tirupati, 9-20 December 2019.